

hw

September 20, 2026

```
[1]: %pip install -q numpy
      %pip install -q matplotlib

import numpy as np
import matplotlib.pyplot as plt
```

Note: you may need to restart the kernel to use updated packages.

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Matplotlib is building the font cache; this may take a moment.

```
[2]: def complex_inner_product(u, v):
      total = 0
      for n in range(len(u)):
          total = total + u[n] * np.conj(v[n])
      return total

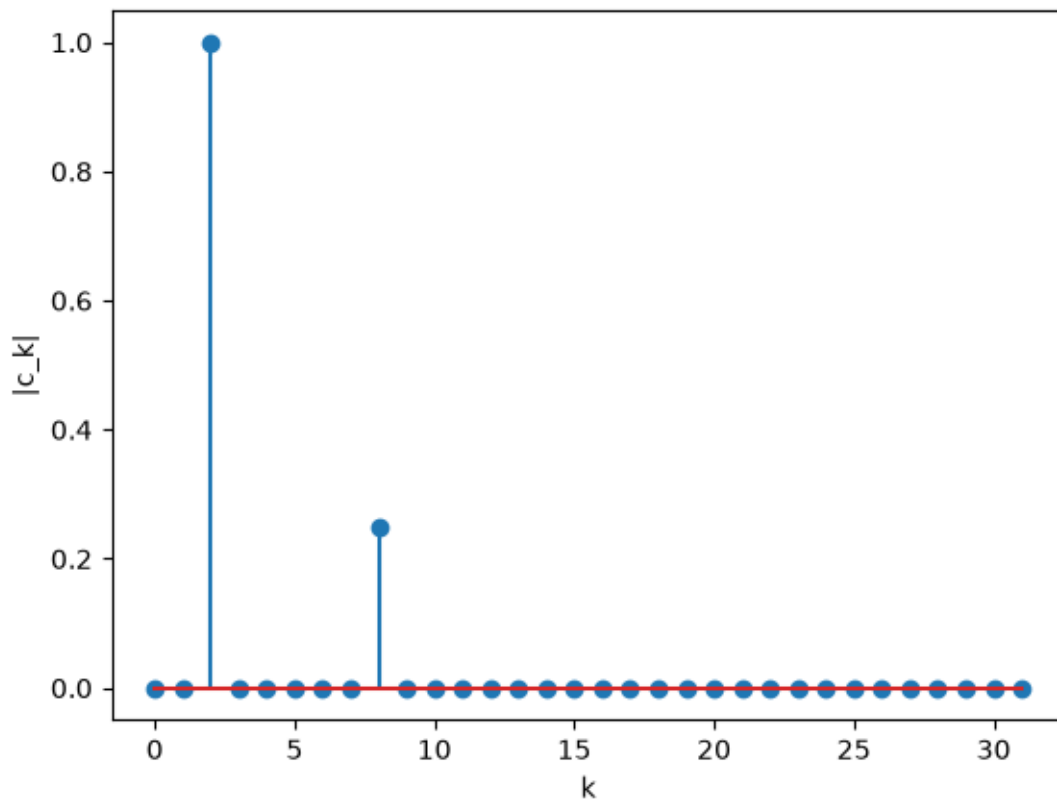
      def generate_basis_vector(k, p):
          w0 = 2 * np.pi / p
          psi = np.zeros(p, dtype=complex)
          for n in range(p):
              psi[n] = np.exp(1j * w0 * n * k)
          return psi
```

```
[3]: def generate_signal(p, frequencies, amplitudes):
      s = np.zeros(p)
      for n in range(p):
          t_n = n / p
          value = 0
          for j in range(len(frequencies)):
              value = value + amplitudes[j] * np.cos(2 * np.pi * frequencies[j] *
↳ t_n)
          s[n] = value
      return s

      s = generate_signal(64, [2, 8], [2.0, 0.5])
```

```
[4]: def get_projection_coefficient(signal, k, p):
      psi_k = generate_basis_vector(k, p)
      return complex_inner_product(signal, psi_k) / complex_inner_product(psi_k,
      ↪ psi_k)
```

```
[5]: mags = []
      for k in range(0, 32):
          mags.append(abs(get_projection_coefficient(s, k, 64)))
      plt.stem(range(32), mags)
      plt.xlabel("k"); plt.ylabel("|c_k|")
      plt.show()
```



```
[6]: s2 = generate_signal(64, [24, 26], [1, 1])
```